

Introduction

Most plastics persist in the environment for centuries and break down into **microplastics** that move through soil, water, and food webs. Earthworms such as *Eisenia fetida* ingest and fragment these particles, potentially altering the chemical properties of their **vermicast** (worm excrement). Increased fragmentation raises **microplastic surface area**, which can enhance chemical absorption and may affect soil fertility. This study tests whether exposure to **expanded polystyrene (EPS)** microplastics changes **nitrite levels** in vermicast.

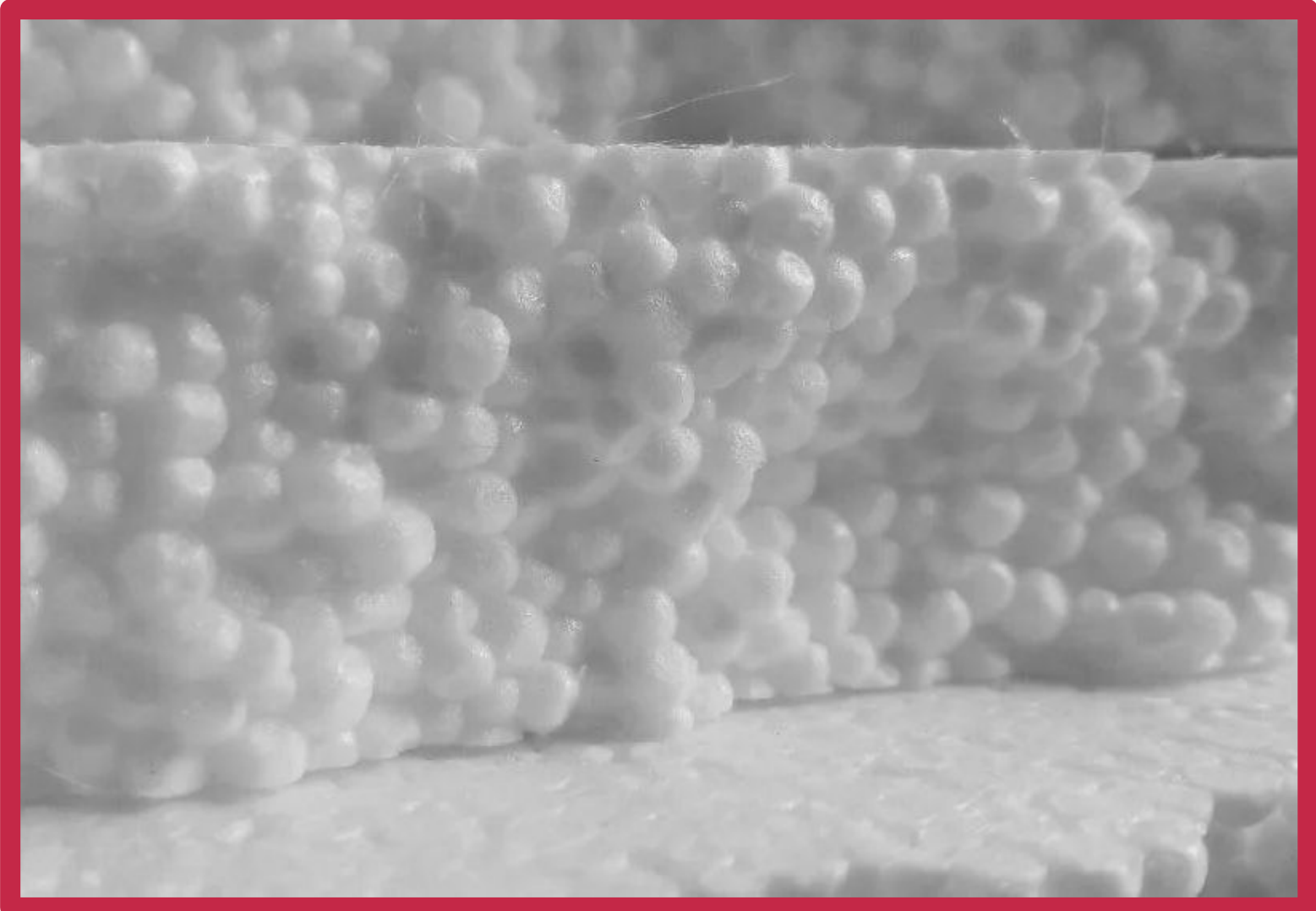


Figure 1. Styrofoam
Note. From How Long Does It Take For Styrofoam To Decompose?, by EPsole, n.d. (<https://epssole.com/how-long-does-it-take-for-styrofoam-to-decompose/>).

Purpose

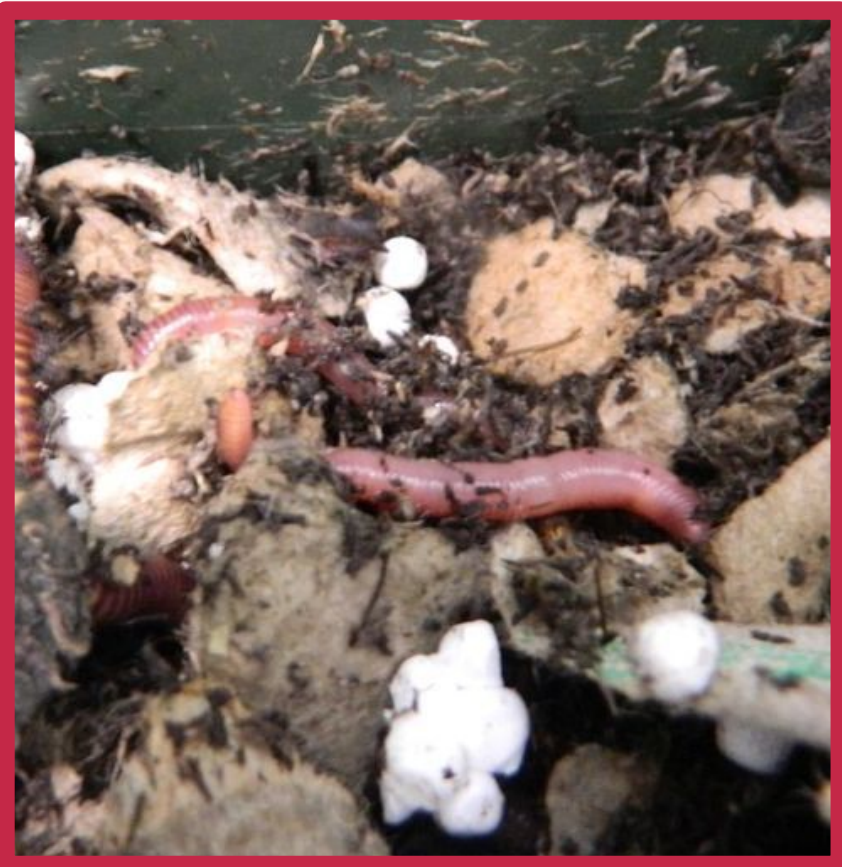
EPS (commonly known as **styrofoam**) is a **low-cost packaging material** that persists in the environment and lacks effective recycling pathways. When ingested by earthworms, EPS microplastics are further fragmented and released in vermicast, potentially altering **microbial processes**. Nitrite is a sensitive indicator of this disruption because it reflects **nitrification activity** and **vermicast quality**. By measuring how EPS microplastics change nitrite levels in vermicast, this study assesses whether a common packaging material could **reduce soil fertility** and **long-term agricultural health**.

Microplastic Fragmentation by *Eisenia fetida* and its Effect on Nitrite Content in Worm Castings

Methodology

Vermicompost & Microplastic Preparation

- Control group (25) with no EPS and test group (25) with EPS
- EPS was blended into microplastics (<5mm)
- Bedding created with ripped egg cartons



Vermicompost Maintenance

- Temperature, humidity, and amount of food was maintained
- Vermicast samples were collected after 12 days



Nitrite Extract & Griess Reaction

- Nitrite was extracted with calcium chloride dihydrate solution then filtered
- Griess reaction converts nitrite to a measurable color



Data Collection

- Vernier VIS Spectrophotometer used to measure nitrite absorbance at 540.1 nm
- Data was organized by group and transferred to DataClassroom

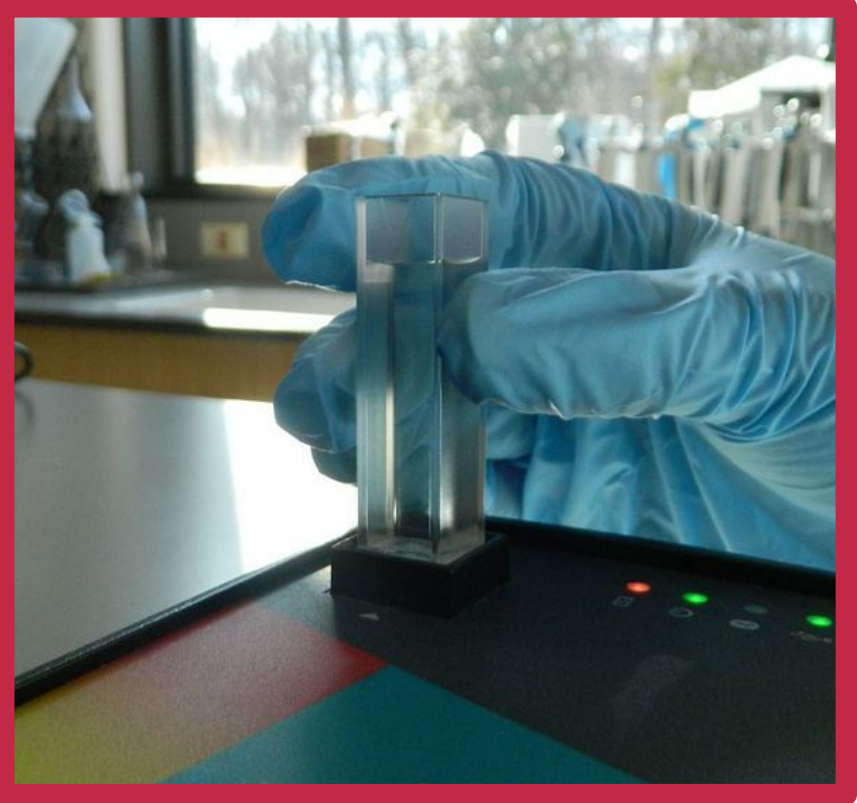


Figure 2. Photo of EPS test group taken by researcher

Figure 3. Photo of frozen watermelon rinds taken by researcher

Figure 4. Photo of nitrite extraction taken by researcher

Figure 5. Photo of cuvette and spectrophotometer taken by researcher

Results

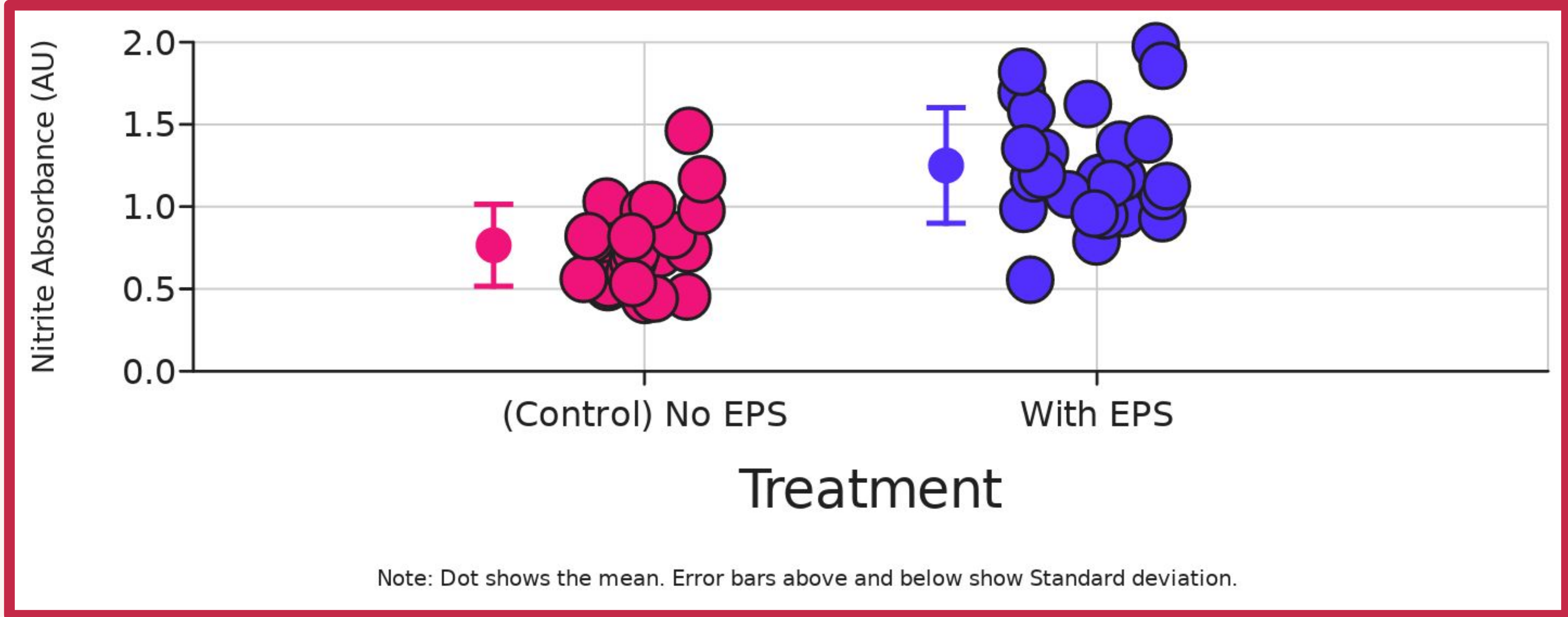


Figure 6. Graph created by researchers using DataClassroom

Group	Mean	Std.dev	SEM	ci95lo	ci95hi	Median	Min	Max	q25	q75
(Control) No EPS	0.766	0.249	0.0498	0.663	0.868	0.743	0.433	1.46	0.535	0.891
With EPS	1.25	0.3512	0.07025	1.105	1.395	1.175	0.556	1.975	0.959	1.409

Figure 7. Table created by researchers using DataClassroom

Conclusion

P-value
<0.01

Interpretation of P
A P-value of <0.01 means extremely strong confidence that the groups are different .

Figure 8. Table created by researchers using DataClassroom

EPS microplastics increased nitrite absorbance in vermicast. The control group averaged **0.766 AU**, while the EPS group averaged **1.25 AU**, a **63% increase** (Figure 7.). The low standard deviation (**0.249 control; 0.3512 EPS**) supports the reliability (Figure 7.). Because nitrite is a sensitive indicator of nitrification, this shift suggests **EPS exposure disrupts nutrient cycling in vermicast**. This may **reduce soil fertility** in microplastic contaminated environments. Future research should examine different microplastic types and concentrations, measure other vermicast nutrients, and longer experiment durations.



Figure 9. An earthworm in soil.
Note. From You Say Worms ... [Photograph], by D. Brekke, 2013, Flickr (<https://www.flickr.com/photos/infospigot/11144693323/>). CC BY 2.0

References

Chen, Yuling, et al. "Defense Responses in Earthworms (*Eisenia Fetida*) Exposed to Low-Density Polyethylene Microplastics in Soils." *Ecotoxicology and Environmental Safety*, vol. 187, Jan. 2020, p. 109788. ScienceDirect, <https://doi.org/10.1016/j.ecoenv.2019.109788>.

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Zhang, S., Pei, L., Zhao, Y., Shan, J., Zheng, X., Xu, G., Sun, Y., & Wang, F. (2023). Effects of microplastics and nitrogen deposition on soil multifunctionality, particularly C and N cycling. *Journal of Hazardous Materials*, 451, 131152–131152. <https://doi.org/10.1016/j.jhazmat.2023.131152>